

Sprint Planning & MVP Communication

SG1 - The Green Mile
Sprint 3

Introduction

This is our group submission for the sprint planning and the MVP communication for Sprint 3.

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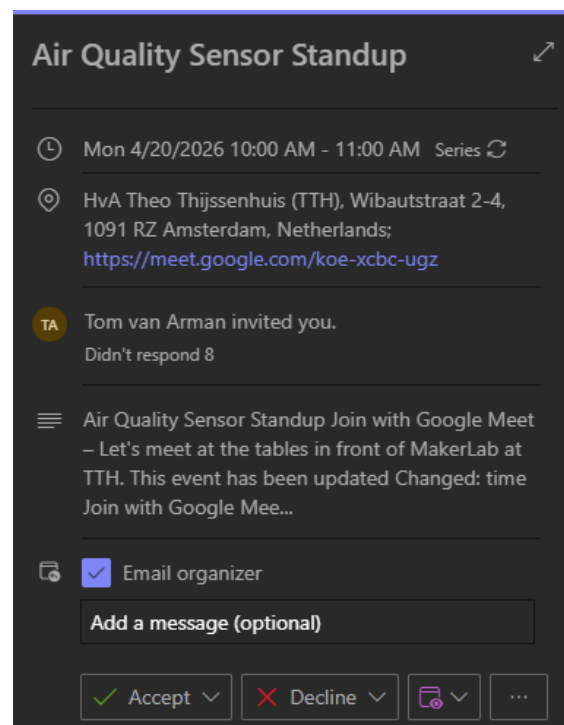
1. MVP Communication

After an intake with our client we came to a conclusion for a sprint goal and a MVP for this sprint:

1. Embedded:
 - Every box to communicate with MQTT
 - PCB designed and ordered
 - Prototype up and running for testing
 - Assemble 4 prototype boxes
 - Fixing sensor misreads and faulty communication
2. Back end:
 - HTTP communication protocol replaced with MQTT to communicate with the back-end
 - JSON payload examples and schema's for the cloud storage team
 - Documentation for setting up own back-end and storage
 - CI/CD
3. MVP:
 - 4 prototype climate boxes (2 SIM card and 2 Wifi boxes)
 - Each box communicates with MQTT
 - Each box has the new internal design
 - At least 2 boxes are ready for field tests

1.1 Meetings with client

We have a meeting with our client each Monday at 10 AM. This will be during the entire sprint. With our MVP meeting being flexible in the third week.



1.2 Feedback from client

PCB (Hardware Design)

- PCB is not required for the current prototypes
- Focus should remain on building and improving working prototypes
- PCB design can be developed in parallel as a separate task
- The Definition of Done includes a print-ready PCB design
- PCB development must not slow down or block prototype progress

Device Naming & Identity

- Replace the current name “CMB” with a clearer and more user-friendly name (e.g. *Airqual*)
- Apply the new naming consistently across all devices and documentation

User Stories Alignment

- Deep sleep improvements and sensor misread handling (assigned to Marnix)
- Ensuring correct and reliable data is sent to the backend (assigned to Hugo)
- Resolve overlap and conflict between these tasks to ensure aligned goals and implementation

Data Quality & Reliability

- Ensure the system sends accurate and trustworthy data
- Reduce incorrect readings and inconsistencies
- Improve stability of the data shown to users

Prototype Design Improvements

- Update the 3D-printed design to include a bracket for the new LED
- Ensure the design remains practical, durable, and easy to assemble

Overall MVP Focus

- Prioritize working, reliable prototypes over perfect hardware design
- Ensure clear structure, consistency, and usability across the system
- Keep development focused on delivering a functional and demonstrable MVP

2. Sprint Planning

The Sprint Backlog is ordered with User Stories (and Epics), highest priorities at the top. The team always selects the top User Story to work on next.

13-apr	11	Sprint 3
20-apr	12	Sprint 3
27-apr	-	Break
4-mei	13	Sprint 3

2.1 Sprint Backlog Requirements

- Weight
- Extensive description
- Acceptance criteria
- Definition of done
- Linked tasks

2.2 Sprint Backlog Evidence

- Screenshot of the Sprint Backlog board

The screenshot displays a Jira Sprint Backlog board for 'Sprint 3'. It is divided into two columns: 'Open' and 'In progress'.

Open Column:

- Item 1:** "As a resident, I want real-time data to assess air quality, so that I know that the quality of air meets national standards in the area the device runs." (Epic, back-end, embedded, #12, 5 votes, 2 To do).
- Item 2:** "As a resident, I want to have multiple devices with different types of internet connections, so that we have data from more areas." (Epic, embedded, #88, 5 votes, 2 To do).
- Item 3:** "As a resident, I want to have 2 devices running on SIM, so that I can test the SIM reliability of the Climate Box in the field." (User Story, #93, 3 votes, 2 To do).
- Item 4:** "As a resident, I want to have 2 devices running on WiFi, so that I can test the WiFi reliability of the Climate Box in the field." (User Story, #92, 2 votes, 1 To do).
- Item 5:** "As a resident, I want the particular matter data to be correct, so that I know for a fact that the data in the dashboard is reliable." (User Story, #91, 2 votes, 1 To do).
- Item 6:** "As a resident, I want an LED to display the state of the setting up part, so that I know at what part I am of setting up the device." (User Story, #25, 3 votes, 1 To do).
- Item 7:** "As a resident, I want the Climate Box to send data every 15 minutes, so that I can have the battery life improved." (User Story, #90, 1 vote, 1 To do).
- Item 8:** "As a resident, I want my air quality readings to be saved in the cloud, so i can always have acces to the readings" (Sub Epic, back-end, embedded, #44, 5 votes, 2 To do).

In progress Column:

- Item 1:** "As a resident, I want a proof-of-concept device to test real-time air-quality monitoring, so I can validate its usefulness before full deployment." (Sub Epic, #16, 5 votes, 4 To do, 1 In progress).
- Item 2:** "As a resident. I want there to be a connection to a cloud storage. So my air quality can get saved for me to see at a later point" (User Story, back-end, #56, 3 votes, 1 In progress).

As a resident, I want my air quality to be saved on cloud storage so i can acces it later anywhere

User Story back-end embedded

#54 3 Sprint 3

1 To do

As a resident, I want a clear and well documented manual that shows how I can register a new CMB box. So that there are no conflicts with the other boxes.

User Story embedded

#87 1 Sprint 3

To do

As a resident, I want to see in a quick eye sight what the current air quality is, so I immediately know if I am breathing good air.

User Story

#94 4 Sprint 3

To do

As a resident, I want the air quality system to use a compact and reliable PCB, so that the device is easy to use, durable, and does not require complex wiring.

Epic embedded rewrite

#85 5 Sprint 3

To do

As a resident, I want the air quality PCB device to measure air quality accurately, so that I can rely on the data in my home.

User Story embedded rewrite

#76 Sprint 3

To do

As a resident, I want the air quality PCB device to work consistently, so that I can depend on it every day.

User Story embedded rewrite

#80 Sprint 3

To do

As a resident, I want the PCB air quality device to work the same as the original system, so that I can trust the results.

User Story embedded rewrite

#84 Sprint 3

To do

- Screenshot of a single Epic

As a resident, I want to have multiple devices with different types of internet connections, so that we have data from more areas.

Open Issue created 16 hours ago by Niklas Kless

Description:

As a resident, I want the devices used in my environment to work reliably with stable connectivity (WiFi or mobile network), so that they function properly in everyday conditions and improvements can be tested without disrupting normal use.

Some devices should be used in real-life conditions, while others are used to test and improve new ideas, ensuring continuous enhancement without affecting regular operation.

Definition of Done:

- ☐ All Blocked by Sub Epics are completed and have been checked.
- ☐ All Blocked by User Stories are completed and have been checked.

0 of 2 checklist items completed - Edited just now by Marnix van der Heijden

0

0

Add design

Create merge request

Child items 0

No child items are currently assigned. Use child items to break down work into smaller parts.

Linked items 2

Blocked by

As a resident, I want to have 2 devices running on WiFi, so that I can test the WiFi reliability of the Climate Box in the field.

#92 2 Sprint 3

User Story

As a resident, I want to have 2 devices running on SIM, so that I can test the SIM reliability of the Climate Box in the field.

#93 3 Sprint 3

User Story

Status

To do

Edit

Assignees

None - assign yourself

Edit

Labels

Epic embedded

Edit

Parent

None

Edit

Weight

5

Edit

Milestone

Sprint 3

Edit

Iteration

None

Edit

Dates

Start: None

Due: None

Edit

Health status

None

Edit

Time tracking

Add an estimate or time spent.

+

2 Participants

Activity

All activity

Oldest first

- Screenshot of a single Sub Epic

As a resident, I want my air quality readings to be saved in the cloud, so i can always have acces to the readings

Open Issue created 2 weeks ago by Thom Koper

Description:

This epic is focused on the whole data storage and pulling for the boxes and the front-end

Acceptance Criteria:

Acceptance criteria based on users requirements.

- ☐ There is a service running that uploads and pulls data from and onto the cloud storage

Definition of Done:

- ☐ All Child items/Tasks are completed and have been checked.
- ☐ All Acceptance criteria are met and have been checked.

0 of 3 checklist items completed · Edited 5 days ago by Thom Koper

👍 0 💬 0 🔒 Add design Create merge request

Child items 0

No child items are currently assigned. Use child items to break down work into smaller parts.

Linked items 3

Blocking

☐ As a resident, I want a dashboard showing real-time air-quality levels so I can immediately see whether local air quality meets national standards.

#17 Sub Epic

Blocked by

☐ As a resident. I want there to be a connection to a cloud storage. So my air quality can get saved for me to see at a later point

#56 3 Sprint 3 In progress User Story back-end

☐ As a resident, I want my air quality to be saved on cloud storage so i can acces it later anywhere

#54 3 Sprint 3 User Story back-end embedded

Status

To do

Assignees

None - assign yourself

Labels

Sub Epic back-end embedded

Parent

None

Weight

5

Milestone

Sprint 3

Iteration

None

Dates

Start: None
Due: None

Health status

None

Time tracking

Add an estimate or time spent.

1 Participant

- Screenshot of a single User Story

As a resident. I want there to be a connection to a cloud storage. So my air quality can get saved for me to see at a later point Edit 🔖 ⋮

Open 📅 Issue created 2 weeks ago by **Thom Koper**

Description:
This userstory focusses on installing and setting up the MQTT broker on the back-end

Acceptance Criteria:
Acceptance criteria based on users requirements.

- ☒ Broker installed
- ☐ Broker has been setup that from anywhere there can be a message sent and recieved
- ☐ HTTP switched to MQTT
- ☐ documentation on how to connect has been made

Definition of Done:

- ☐ All **Child items/Tasks** are completed and have been checked.
- ☐ All **Acceptance criteria** are met and have been checked.

1 of 6 checklist items completed · Edited 3 hours ago by Thom Koper

👍 0 👎 0 🗨️ Add design Create merge request ⌵

Child items 4 🔍 + ⌵

- ☒ Create documentation and visuals for the team to connect to the back-end via the new MQTT To do ×
#71 🔗 Sprint 3 Task back-end
- ☒ Replace old HTTP connection with the new MQTT connection To do ×
#70 🔗 Sprint 3 Task back-end
- ☒ Setup MQTT Broker To do ×
#53 Task back-end
- ☒ Install the MQTT broker Done Closed ×
#52 In progress Task back-end

Linked items 1 🔍 + ⌵

Blocking

- ☐ As a resident, I want my air quality readings to be saved in the cloud, so i can always have acces to the readings 2 👍 👎 1 To do ×
#44 🔗 5 🔗 Sprint 3 Sub Epic back-end embedded

Metadata:

- Status:** In progress Edit
- Assignees:** Thom Koper Edit
- Labels:** In progress × User Story × back-end × Edit
- Parent:** None Edit
- Weight:** 3 Edit
- Milestone:** Sprint 3 Edit
- Iteration:** None Edit
- Dates:** Start: None Due: None Edit
- Health status:** None Edit
- Time tracking:** Add an estimate or time spent. +
- 1 Participant:**

- Direct link to Sprint Backlog in Gitlab

https://gitlab.fdmci.hva.nl/studio/smart-cities/projecten/2025-2026-semester-2/group-project/sg1-the-green-mile/-/boards/41518?milestone_title=Sprint%203